

Principles of Data Science Project:

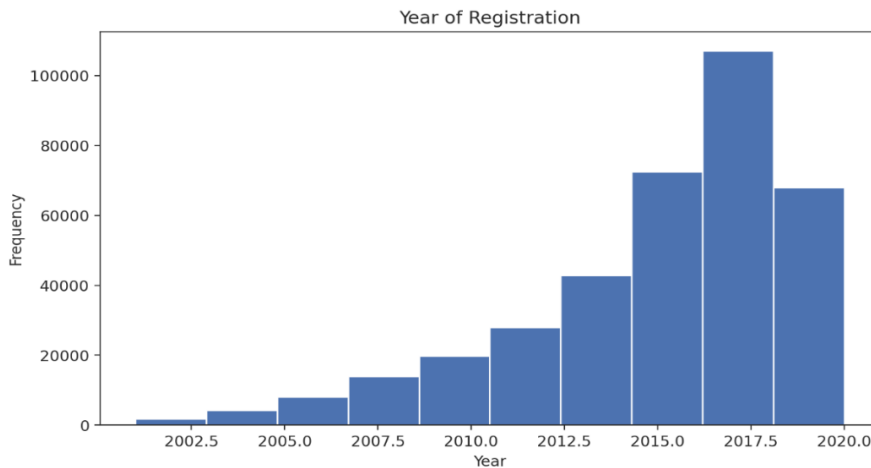
1. Data Understanding and Exploration

1.1 Meaning and Type of Features (3-4)

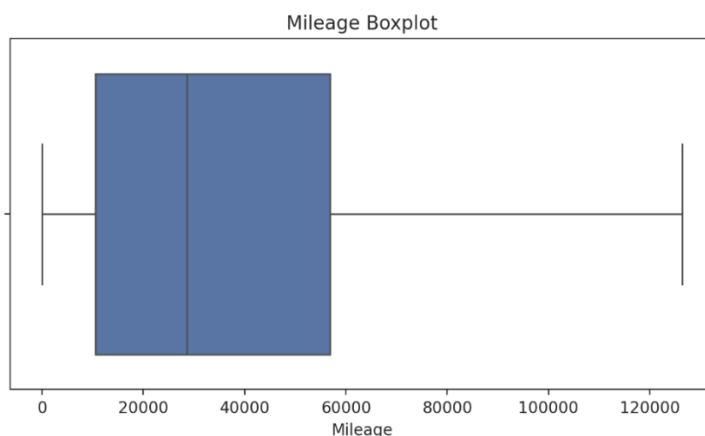
1. The **year_of_registration** column consists of 84 unique numeric values. It is quantitative and is stored as a float data type to one decimal place. This column indicates the year in which the vehicles were registered and can be used to ascertain the vehicle age. It could be a good indicator of the price of the vehicle.
2. The **standard_make** column indicates the vehicle manufacturer and consists of 110 different makes. The values are categorical and are recorded as object value data which indicates the presence of non-numerical values. This data may be a useful indicator of price although manufacturers may offer a variety of models at different price ranges.
3. The values in the **reg_code** column represent information from the vehicle license plate. The values are stored as object data type. There are 72 unique values including non-numeric as well as the numeric values. From this column we should be able to ascertain the year of registration and hence the age of the vehicle. In this respect it serves the same purpose as the year_of_registration column. In 2001, the current rules on the way license plates display information was set out. This replaced the system which had run from 1983 to 2001.
4. The **mileage** column has numerical values to one decimal place stored as float data type. The values are quantitative and represent how many miles the vehicle has travelled and is likely to influence the price of the vehicle.

1.2 Analysis of Distributions (3-4)

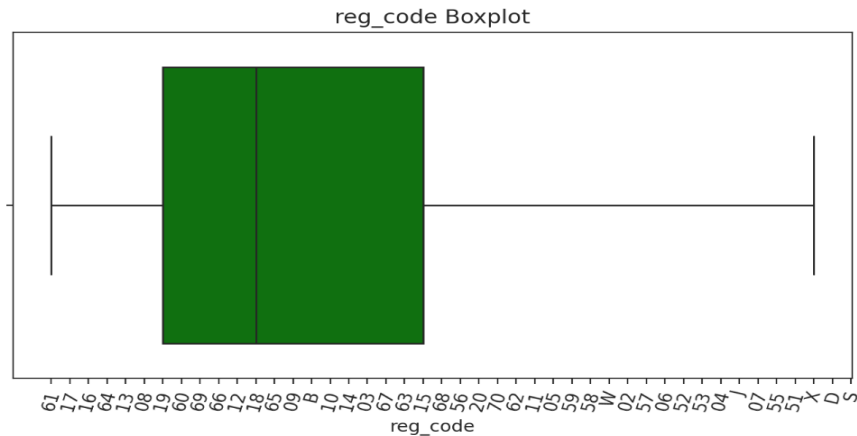
1. The values in the 'year of registration' column range from 1999 to 2020. Over 99% of the available data, lies in the period from 2001 to 2020. The top 50% of the data falls between 2016 and 2020. The mean year is 2015, although this will be affected by potentially erroneous values. There are 33311 missing values. A histogram of the values from 2001 onwards shows that our data is negatively skewed with a mode year of 2017.



2. The mean mileage value is just short of 38,000 miles and over 75% of the values are between 0 and 60,000 miles. The maximum value is 999,999 miles. A boxplot of the mileage which ignores the outliers sets the upper bound at just over 120,000 miles. It shows the values are positively skewed. This indicates a small quantity of very large values and is consistent with the large standard deviation showing the values are spread over a wide range.



3. The reg code column has a mode value of 17 indicating the year 2017 which is consistent with the findings of the year of registration column. Before 2001 the year of registration was indicated by a letter starting from A in 1963, B for 1964 and so on. In 1983 the system restarted at A. The letters I, O, Q, U and Z were not used as an indicator of the year. By looking at a boxplot with the outliers removed, we find most of the data corresponds to the rules for 2001 onwards and has positive skew. There is a large chunk of missing values, more so than in the year of registration column.



4. The standard_colour column has 22 unique values, 5378 null values. The mode value is Black and the 6 most frequently occurring colours make up over 93% of the available data.

standard_colour	
count	396627
unique	22
top	Black
freq	86287

```
#Examine frequency of most common colours.
top_6_colour = advt['standard_colour'].value_counts().nlargest(6)
print(f'The 6 most popular colours make up {top_6_colour.sum()/len(advt)*100:.2f}% of the data')
```

The 6 most popular colours make up 93.58% of the data

2. Data Pre-Processing

2.1 Data Cleaning (e.g., dealing with incorrect values, outliers) (2-3)

1. The year_of_registration column has 33311 null values out of 402005 which is just over 8% of the data. We find that over 90% of these null values, (31249/33311) are from vehicles that have low mileage and condition new. Generally, vehicles with a mileage of under 200 are considered new. Hence, we assign this missing data with the year of 2020, the maximum year, being the most recent year included in the database.

```
# If vehicle condition is new and mileage < 200, we replace year of registration null values with the year of registration max value.
max_year = advt['year_of_registration'].max()

advt.loc[(advt['vehicle_condition'] == 'NEW') &
         (advt['year_of_registration'].isnull()) &
         (advt['mileage'] < 200), 'year_of_registration'] = max_year
```

For the remaining 2062 null-values we can use the reg code data where available to approximate the year. Firstly, we need to convert the reg code column to float type.

```
advt['reg_code'] = pd.to_numeric(advt['reg_code'], errors='coerce')
```

```
[28] advt.loc[(advt['reg_code'] <= max_year-2000)
          & (advt['year_of_registration'].isnull()), 'year_of_registration'] = 2000 + advt['reg_code']
```

```
[29] advt.loc[((advt['reg_code'] > 50) & (advt['reg_code'] < 50 + (max_year-2000)))
          & (advt['year_of_registration'].isnull()), 'year_of_registration'] = 2001 - 50 + advt['reg_code']
```

That leaves just 584 null-values. We could replace them with mean year. On examination we see the mean price for these values is much higher than for the general data. Therefore, we might create incorrect data if for example we assign the mean year to a vehicle that was out of production by then or that has much larger mileage or price than is typical. We can afford to ignore such a tiny percentage of the data set.

2.Outliers: Since the registration system began in 1903, any values prior to that are likely to be errors. We can use the data from the reg_code column to address these errors and replace them with the correct year using code similar to that already used. Since we will no longer use the reg_code column, we remove it from the data frame.

3.The mileage has very high outlier values. From the boxplot we have the upper bound as roughly 125000m and on investigation we find less than 1% of the mileage is above 150000m and the vehicles that have these values are mostly over 10 years old and so likely to have high mileage. We can replace with these large mileage values with a limit of ($Q3+2*IQR=149665$) to ease visualisation.

```
# if mileage > Q3+2*IQR, we replace with Q3 + 2xIQR

Q1 = advt['mileage'].quantile(0.25)
Q3 = advt['mileage'].quantile(0.75)
IQR = Q3 - Q1

upper_bound = Q3 + 2 * IQR
advt.loc[advt['mileage'] > upper_bound, 'mileage'] = upper_bound
```

The remaining 127 mileage null values we replace with the median mileage with awareness that some inaccurate data may be created such as vehicles with a year of 2020 are unlikely to have the median mileage.

```
# replace mileage null values with median
median_mileage = advt['mileage'].median()
advt['mileage'].fillna(median_mileage, inplace=True)
advt.info()
```

We could replace the null values in the fuel_type and body_type columns with the relevant average value but we risk creating incorrect data unless we investigate further. Since each column has less than 0.5% null values, on this occasion we will ignore them.

2.2 Feature Engineering (e.g., deriving informative features) (2-3)

1. We create a new column showing the age, in years, of the vehicles in our data set. This column will make it easier to see the age of the vehicle.

```
# Feature Engineering: Create age column

max_year = advt['year_of_registration'].max()
advt['age(in years)'] = (max_year - advt['year_of_registration'])
```

2. Average_Mileage: We create a new feature showing mileage per annum in years taking care not to divide by zero for the new (max year) vehicles. This will help distinguish

```
# New column average_mileage with values of mileage/(year >=1)

# Calculate 'mileage_per_year'
advt['average_mileage'] = advt['mileage'] / (advt['age(in years)'] + (advt['age(in years)'] == 0))
advt['average_mileage'] = advt['average_mileage'].round(1)
```

between vehicles that have done a lot of miles in a short space of time and those that have a high mileage but comparably low mileage per year.

3. We categorize the average mileage column into low, medium and high based on the quartiles of the mileage data. This creates a helpful new feature.

```
# Engineer categories for average mileage column.

# Find quartiles
Q1_mileage = advt['average_mileage'].quantile(0.25)
Q3_mileage = advt['average_mileage'].quantile(0.75)

# Create a new column
advt['mileage_class'] = 'medium' # Default category

# Assign categories based on quartiles
advt.loc[advt['average_mileage'] < Q1_mileage, 'mileage_class'] = 'low'
advt.loc[advt['average_mileage'] > Q3_mileage, 'mileage_class'] = 'high'
```

2.3 Subsetting (e.g., filtering, feature selection, and row sampling) (2-3)

1 Two of the most expensive vehicles in the data are a yellow Lamborghini Urus and a green Aston Martin vanquish. Alternatively, we can look at the most common 5 makes which comprise over 40% of the data frame. Of these 5 makes, the 3 with the highest mean prices are Mercedes Benz, Audi and BMW with mean prices of around £20,000. However, there might be considerable variation in the price of the models available for these makes.

```
#list mean price of top 5 makes
top_make = advt['standard_make'].value_counts().nlargest(5).index
filtered_advt = advt[advt['standard_make'].isin(top_make)]
filtered_advt['standard_make'].value_counts()
average_prices = filtered_advt.groupby('standard_make')['price'].mean().sort_values(ascending=False)
average_prices.round(2)
```

price	
standard_make	
Mercedes-Benz	21524.05
Audi	20419.64
BMW	19916.59
Volkswagen	14011.70
Vauxhall	8178.43

2.The make that occurs most frequently is BMW. The BMW model with the highest price is the Z8 convertible with prices ranging between £150,000 and £200,000. The average BMW available is roughly 5 years old and priced around £20,000 which is slighter higher than the mean price for all the vehicles.

3. If we group the hybrid vehicles together, we see the mean price is just over £25,500 which is higher than the average for the wider dataset. The average age is 2-3 years which is lower than the overall average. The most common make on our data frame is Toyota and the the oldest hybrid vehicle we have is a 17-year-old silver Toyota Alphard MPV.

```
# We create a subset for rows where 'fuel_type' contains 'hybrid' (case-insensitive)
hybrid_cars = advt[advt['fuel_type'].str.contains('hybrid', case=False, na=False)]
hybrid_cars.describe().round(2)
```

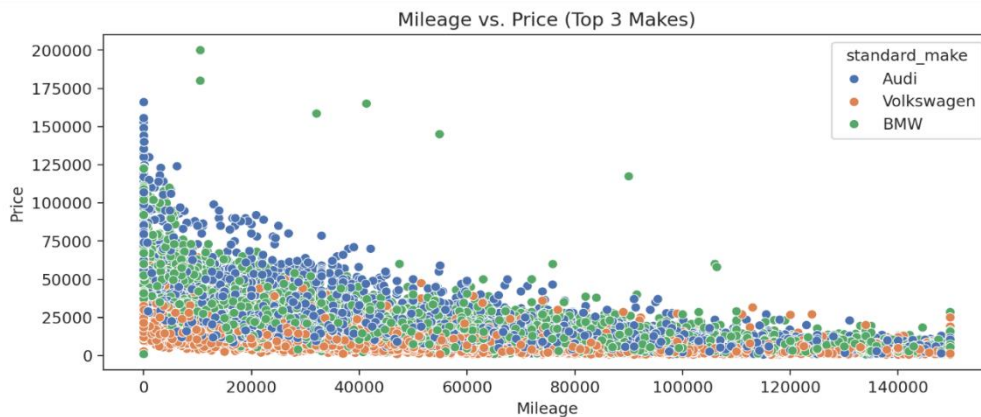
```
hybrid_cars['standard_make'].value_counts().head(5)
```

count	
standard_make	
Toyota	7015
Lexus	2547
Volvo	2479
BMW	1933
Mitsubishi	1100

3. Analysis of Associations and Group Differences

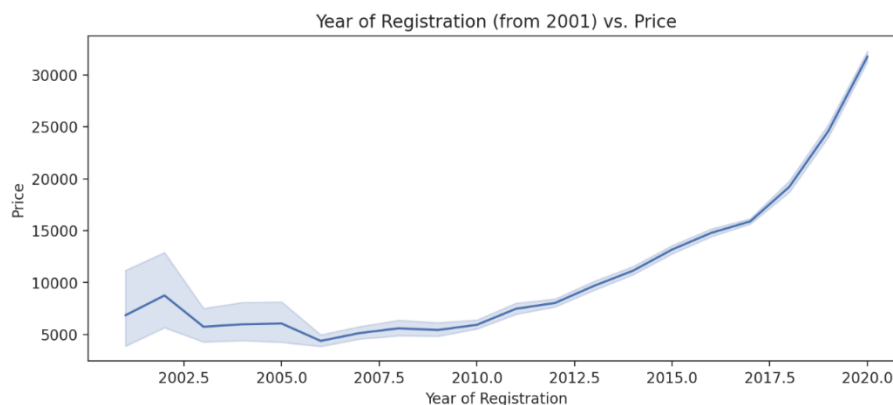
3.1 Quantitative – Quantitative (2-3)

Mileage vs price using a data sample of 50,000 and winsorizing the price for ease of visualization.



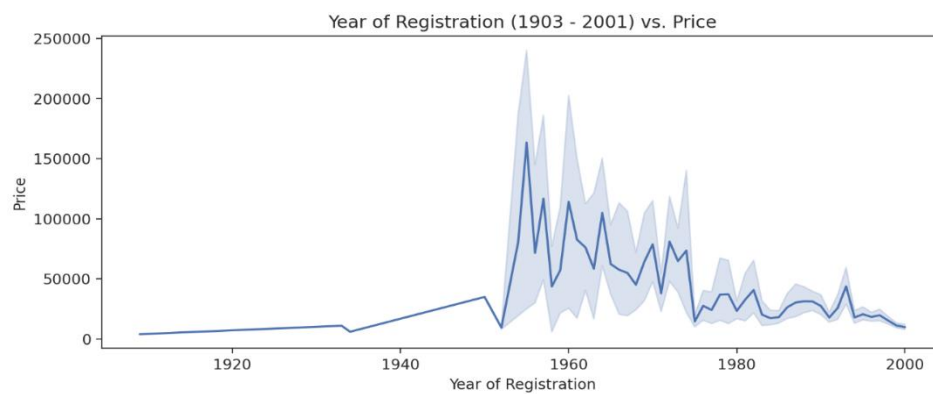
For the most common 3 makes the value of the vehicle tends to decrease linearly as the mileage increases, which is consistent with the wider dataset.

2. Year of registration from 2001 onwards against price, using sample of 50,000.



From 2020 we see an initial rapid decline in value of vehicles to 2017. The price then decreases at a lower rate from 2017 to 2006. From 2006 to 2001 the price seems inconsistent with the general trend, briefly rising before falling and warrants further investigation.

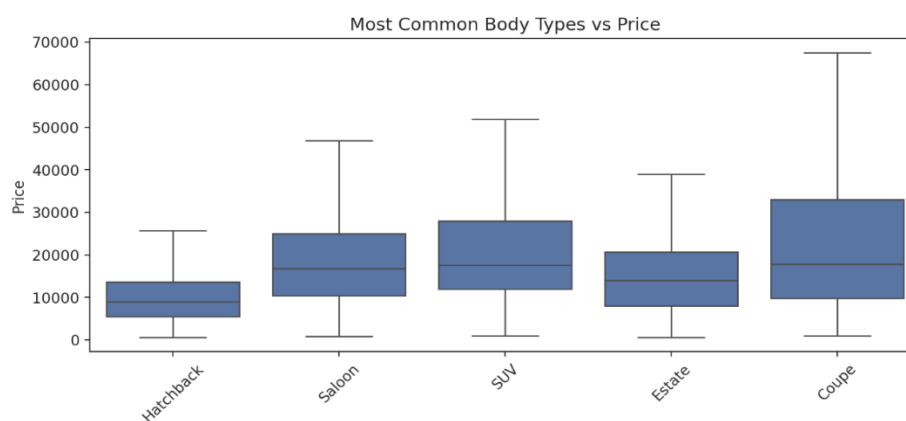
3.



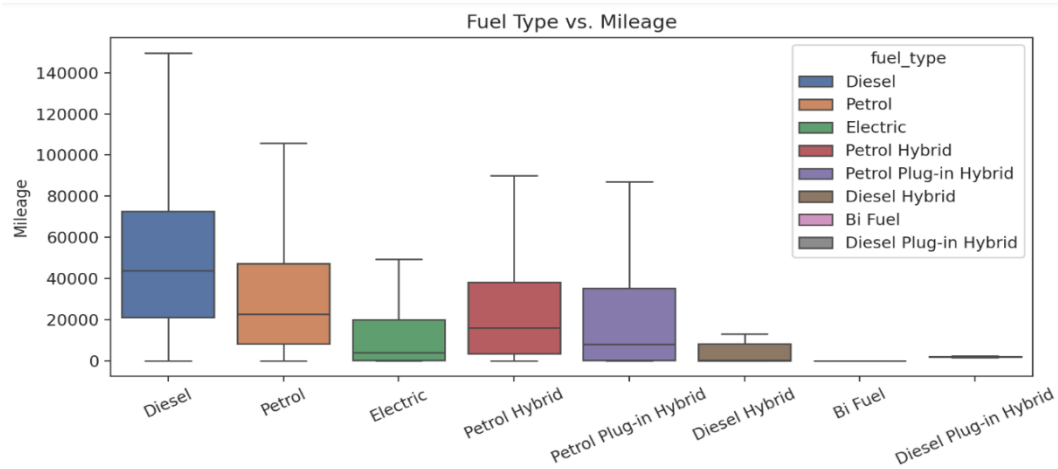
For the period 1903 to 2001 the trend is more erratic. Since this is a small dataset it is more influenced by individual variation. It does show a general decline in prices from 1950 to 2000. This could be explained by older vehicles being regarded as collectables if well maintained.

3.2 Quantitative - Categorical (2-3)

1. Price against top 5 most common body type. Hatchbacks occurred most in the dataset but seem to be associated with a lower price range in comparison to other body types.

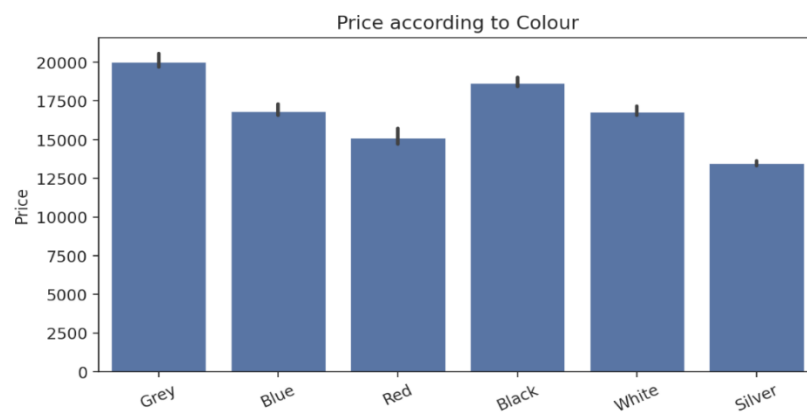


2.mileage vs fuel type



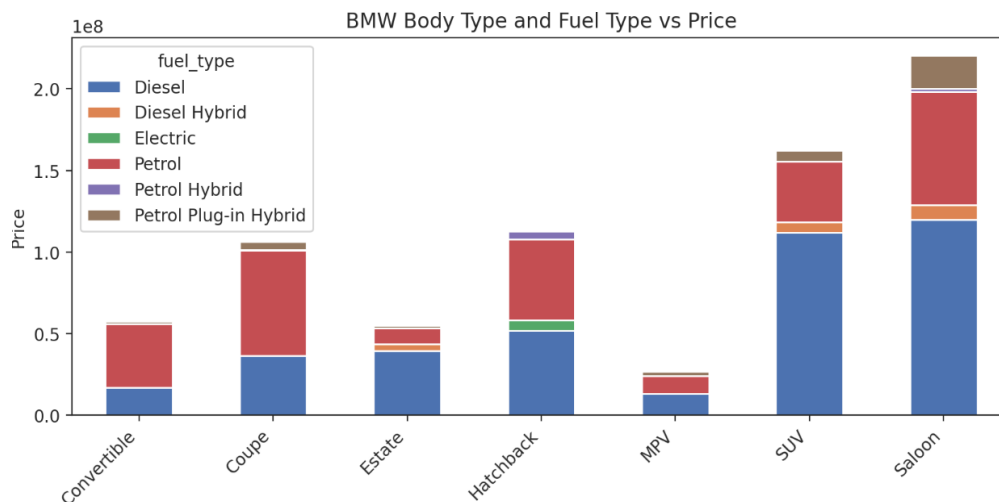
Diesel vehicles tend to have done the most mileage, followed by petrol vehicles. Unsurprisingly modern fuel types such as electric are at the lower end of the mileage range.

3.The bar chart below suggests there is a slight difference in the most common 6 colours of vehicle in relation to price, with grey emerging as the overall winner and black in second place which is unexpected since black is the mode colour.

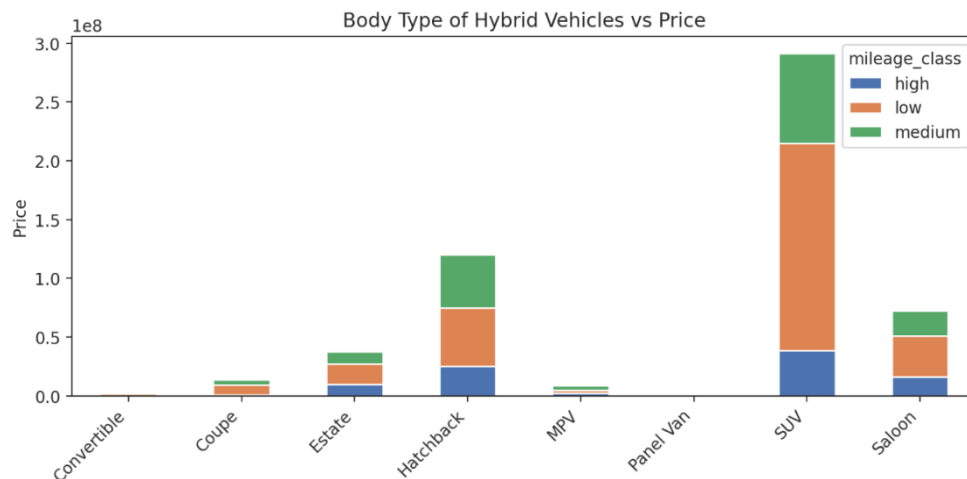


3.3 Categorical – Categorical (2-3)

1. BMW fuel and body type vs price: Looking solely at BMW vehicles in our data we see most are either petrol or diesel, which is consistent with the wider dataset and that the sum price of the saloon vehicles is greatest. Most convertibles are petrol, and most SUVs are diesel. The hatchback has the most noticeable component of electric engines, and the saloon noticeably has hybrid options.



2. Body types of hybrid vehicles vs price with mileage class.



We see most of the hybrid vehicles are low or medium mileage. The greatest sum of price by body type is the SUV which is also the most common hybrid body type.

Google Colab and its generative AI features were used during the making of this report.